



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam Summer 2023

CSE 4611/CSI 411: Compiler Design/Compiler

Total Marks: 40 Duration: 2 hours

Answer all questions. Figures in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1.	Read the following code and detect different types of errors. <pre>1. include<stdio.h> 2. include<canio.h> 3. void foo (int a, void b; float c){ 4. if(a/c = 0) 5. print("Done"); 6. return a; 7. } 8. int main(){ 9. int a1 = 0, b1 = 1, 1a; 10. flaot c, temp = 0; 11. a = b1 + c; 12. temp = fao(b1, a); 13. a1 = a1>b1 ? "a" : "2" ; 14. if(a1=b1) 15. foo(a, c1); 16. else(temp == foo(a1, b1, c)) 17. printf("Not Done : %d", a); 18. return c; 19.}</pre>	8
2.	Construct a Predictive Parsing table for the following grammar & also check whether string "cdbcaadbc" is accepted or not. The sequence of Moves by Predictive Parser in a table (using stack). $A \rightarrow Bc / Aa / c$ $B \rightarrow Bb / Ad / d$	[4]

3.	Eliminate immediate left recursion from the following grammar.	[4+4]
	(a) $E \rightarrow E+E \mid (E) \mid EA \mid *A$ $A \rightarrow AB \mid AAC \mid Ba \mid Cb \mid \epsilon$ $B \rightarrow B + B \mid BBAD \mid B(BA) \mid b \mid \epsilon$ $D \rightarrow DC \mid D/C \mid d/C$ $C \rightarrow (C) \mid a \mid b \mid c \mid E$ b) $M \rightarrow MooN \mid mN \mid MN \mid \epsilon$ $N \rightarrow Night \mid NN \mid Noon \mid \epsilon$ $O \rightarrow OP \mid oOP \mid ON \mid Pot \mid \epsilon$ $T \rightarrow 1 \mid 2 \mid 3 \mid \epsilon$	
4.	Left Factor the following. (a) Consider each word as a single terminal. $X \rightarrow \text{Poet Loves Poetry} \mid \text{Poets eat Sugar} \mid \text{Parrot talks} \mid \text{palm tree} \mid \text{Panache} \mid \text{Poker face} \mid \text{polite nature} \mid \text{polishing} \mid \text{Poetic} \mid \text{police chase} \mid \text{parochial} \mid \text{paradox} \mid \text{persisting}$	[4+4]
	(b) Consider all the terminals are of single character/number/symbol. $A \rightarrow ([2+3]) \mid ((a+b)) \mid (2+3(4*5)) \mid (2-3(1/5)) \mid (2*3) \mid (2+5(1*2)) \mid (2+3(1/2)) \mid ((a+4)) \mid ((2*6))$ $B \rightarrow \text{animal} \mid \text{anemic} \mid \text{anatomy} \mid \text{analytical} \mid \text{archaic} \mid \text{arcane} \mid \text{ardor} \mid \text{anapol} \mid \text{aneche} \mid \text{arcagi}$	
5.	For the following grammars, write down the FIRST and FOLLOW sets for each nonterminal. Consider all the terminals are of single character/number/symbol.	[4+4+4]
(a)	$X \rightarrow eXYTgh \mid iXPrq \mid YtPrq \mid Pgh \mid cd$ $Y \rightarrow pqXP \mid c \mid d \mid Piy \mid TroXg \mid \epsilon$ $P \rightarrow Yq \mid PcTX \mid iqc fXiyYqXY \mid cY \mid \epsilon$ $T \rightarrow e \mid u \mid YPjl \mid iYrTP \mid \epsilon$	(c) $I \rightarrow L + Dc \mid IopL \mid IqDQL + \mid QLI5c \mid \epsilon$ $L \rightarrow abIch \mid agIrx \mid aQD - Li \mid ILbQD \mid \epsilon$ $D \rightarrow Qst \mid Llb \mid IL(a) \mid D + LQo \mid De \mid \epsilon$ $Q \rightarrow aIQa \mid QrtLI d \mid QxyDL \mid cD Io \mid \epsilon$
(b)	$P \rightarrow iPRtG \mid abGi \mid R \mid GRc \mid xy$ $R \rightarrow abcRG \mid cPG \mid io \mid cR \backslash t \mid \epsilon$ $G \rightarrow pGRc \mid y * \mid xR + y \mid aPR \mid \epsilon$	